SURVEYING & LEVELLING · NOTE SL01

Surveying — Definition, Principles, Kinds of Survey and Scales

Course topics: SWCE1.4 L1, SWCE1.4 L2

What surveying is

Surveying is the science of measuring lines, distances, angles and elevations on or near the earth's surface to fix locations and directions and to work out areas and volumes.

Part of the work	What you do
Field work	Take the measurements on the ground
Office work	Compute and draw
Data representation	Present the results as maps, plans, profiles and cross-sections

Two general principles (ICAR eCourse)

1. **Work from the whole to the part.** Enclose the area with main stations and main survey lines first, then split it into well-conditioned triangles. Errors then stay local instead of accumulating.
2. **Fix every new station by at least two measurements** (linear or angular) from fixed reference points.

Plane and geodetic surveying

Basis	Geodetic surveying	Plane surveying
Curvature of the earth	Taken into account	Neglected — the surface is treated as a plane
Area covered	Large areas	Small areas only
Angles	Spherical	Treated as plane angles
Used for	Precise positions of widely distant points	Roads, canals, buildings, irrigation and drainage works

Kinds of surveys

Survey	Purpose
Cadastral (land) survey	Fix land boundaries and areas
Topographic survey	Record the shape (relief) and the natural and artificial features
Agricultural survey	Contours, terrace lines, land levelling, irrigation and drainage works, farm maps
Construction (engineering) survey	Lay out lines, grades and dimensions for engineering works
Route survey	Railways, roads, canals, drains and power transmission lines
Aerial survey	Cover large areas with photographs
Photogrammetric survey	Take measurements from photographs

Aerial survey and photogrammetry

- Photograph the ground from an aircraft; the pictures support project planning.
- Take measurements directly from single photographs or from stereo pairs.
- The process of measuring from photographs is **photogrammetry**.

Scales

A scale is the fixed relationship between a distance on the map and the matching distance on the ground. Express a map scale as a **representative fraction (RF)**:

$$\text{RF} = \text{length on drawing} / \text{corresponding length on ground}$$

- RF has no units, so it does not depend on the unit system.
- **Numerical scale** — written as an RF; it becomes wrong if the paper shrinks or expands.
- **Graphical scale** — drawn on the map; it shrinks and stretches with the paper, so prefer it for measuring distances off the sheet.
- Do not choose a larger scale than the job needs. Field maps usually range from 1 cm = 5 m to 1 cm = 50 m.

Construct a scale of 1 cm = 3 m that can read 38 m.
1. Draw a 20 cm line — it represents 60 m.
2. Divide it into 6 equal parts; each part is 10 m.
3. Subdivide the first part into 10 parts; each small part is 1 m.
4. Read 38 m as three main parts (30 m) plus 8 small parts (8 m).

RF = 1 cm / 3 m = 1/300.

Sources: ICAR eCourse 'Surveying and Leveling' — course content by Dr D. Tamil Mani (Dept. of Soil & Water Conservation Engg., AEC&RI, TNAU Coimbatore), reviewed by Dr V. M. Abdul Hakkim (KCAET, Kerala Agricultural University); published as an AgriMoon e-book. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Surveying in which the curvature of the earth is neglected is called

(a) Geodetic surveying
(b) Plane surveying
(c) Hydrographic surveying
(d) Aerial surveying

Q2. The representative fraction of a scale 1 cm = 3 m is

(a) 1/30
(b) 1/300
(c) 1/3000
(d) 3/100

Q3. Which scale stays correct even if the map paper shrinks?

(a) Numerical scale
(b) Representative fraction
(c) Graphical scale
(d) Diagonal scale only

Q4. A survey made to fix land boundaries and areas is a

(a) Topographic survey
(b) Route survey
(c) Cadastral survey
(d) Photogrammetric survey

Q5. The second general principle of surveying says a new station should be fixed by at least

(a) one measurement
(b) two measurements
(c) three angles
(d) four offsets

Answers

Q1 — (b) Plane surveying — *Plane surveying treats the surface as a plane and suits small areas.*

Q2 — (b) 1/300 — *3 m = 300 cm, so RF = 1/300.*

Q3 — (c) Graphical scale — *A graphical scale is drawn on the paper and shrinks with it.*

Q4 — (c) Cadastral survey

Q5 — (b) two measurements — *ICAR eCourse: at least two linear or angular measurements from fixed points.*